Moments of Points, Line Segments, Polygons, and Polyhedra

A Step-by-Step Derivation in 3D (Undergraduate-Friendly)

Chunk 1: Theory + Closed Forms (No Code Yet)

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1 Why moments in 3D, and what we will derive

Moments are a powerful and compact way to represent the shape of a geometric object. The 2D version of this framework covers points, line segments, and polygons. This document derives the *3D extension*: we add a third spatial dimension z to every shape and formula, and introduce the polyhedron (solid 3D body) as the natural 3D counterpart of the 2D polygon. We also keep the 3D *flat polygon* as an explicit case: it is a planar 2D region embedded in 3D space, and its moments are surface integrals rather than volume integrals.

We derive closed-form moment equations for four kinds of shapes:

- **Point** — all mass at one location in $\mathbb{R}^3$,
- **Line segment** — a 1D curve in $\mathbb{R}^3$,
- **Polygon** — a flat planar region embedded in $\mathbb{R}^3$ (surface integral),
- **Polyhedron** — a solid 3D region bounded by planar faces (volume integral).

and for three popular basis families:

- **Geometric (monomial)** — $x^m y^n z^p$,
- **Fourier (complex exponential)** — $e^{-\mathrm{j}\pi(mx+ny+pz)}$,
- **Legendre (orthogonal polynomial)** — $P_m(x)P_n(y)P_p(z)$.

That gives $4 \times 3 = 12$ cases in total.

Important promise. Every boxed equation in this document is a true closed form: no remaining unevaluated integrals, only finite sums and elementary functions (powers, exponentials, factorials), and expressions that map directly to code.

Coordinate domain. We assume the geometry has been normalized so all coordinates lie in $[-1, 1]^3$. (Apply an affine transform to your data first if needed.)

2 What is a moment?

Let $f(x, y, z)$ represent a shape in 3D space. A moment coefficient C_{mnp} indexed by non-negative integers m, n, p is:

$$C_{mnp} = \int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} f(x, y, z)\, \phi_{mnp}(x, y, z)\, \mathrm{d}x\, \mathrm{d}y\, \mathrm{d}z. \tag{1}$$

Think of this as a dot product $C_{mnp} = \langle f, \phi_{mnp} \rangle$. Low-order indices capture coarse global properties; high-order indices capture fine local detail.

3 The three bases (fully defined)

3.1 Geometric (monomial) basis

$$\phi^{\mathrm{geo}}_{mnp}(x, y, z) = x^m y^n z^p. \tag{2}$$

Simple to compute; not orthogonal.

3.2 Fourier basis

Let $\mathrm{j} = \sqrt{-1}$.

$$\phi^{\mathrm{F}}_{mnp}(x, y, z) = e^{-\mathrm{j}\pi(mx+ny+pz)}. \tag{3}$$

The coefficients are generally complex-valued. They are the 3D Fourier transform evaluated at integer frequency triple (m, n, p).

3.3 Legendre basis

$$\phi^{\mathrm{L}}_{mnp}(x,y,z) = P_m(x)\,P_n(y)\,P_p(z), \tag{4}$$

where P_m is the Legendre polynomial of degree m on $[-1,1]$:

$$P_m(x) = \sum_{k=0}^{\lfloor m/2 \rfloor} A_{m,k}\,x^{m-2k}, \qquad A_{m,k} = (-1)^k \frac{(2m-2k)!}{2^m\,k!\,(m-k)!\,(m-2k)!}. \tag{5}$$

Sanity Check 3.1. $P_0(x) = 1$, $P_1(x) = x$, $P_2(x) = \frac{1}{2}(3x^2-1)$. All match the standard Legendre polynomials.

4 Shape models (point, segment, polygon, polyhedron)

4.1 Point model

A point at (x_0, y_0, z_0) is represented by three Dirac deltas:

$$f_{\mathrm{pt}}(x,y,z) = \delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0). \tag{6}$$

4.2 Line segment model

Let the segment start at $\mathbf{a} = (a_x, a_y, a_z)$ and end at $\mathbf{b} = (b_x, b_y, b_z)$. Define the displacement $\mathbf{d} = \mathbf{b} - \mathbf{a} = (d_x, d_y, d_z)$ and length $L = \|\mathbf{d}\|$. Parameterize by $t \in [0,1]$:

$$x(t) = a_x + td_x, \quad y(t) = a_y + td_y, \quad z(t) = a_z + td_z.$$

The arclength-weighted model is:

$$f_{\mathrm{seg}}(x,y,z) = \int_0^1 \delta(x-x(t))\,\delta(y-y(t))\,\delta(z-z(t))\,L\,\mathrm{d}t. \tag{7}$$

The factor L converts the parameter measure to arclength ($\mathrm{d}s = L\ \mathrm{d}t$).

4.3 Polygon model (flat planar region in $\mathbb{R}^3$)

A *3D polygon* is a flat, filled, planar region $\Sigma \subset \mathbb{R}^3$. It has a well-defined unit normal $\hat{\mathbf{n}}_\Sigma$ and a finite area. We represent it using a surface delta:

$$f_{\mathrm{pgon}}(x,y,z) = \iint_\Sigma \delta^{(3)}\big((x,y,z) - \mathbf{r}\big)\,\mathrm{d}S(\mathbf{r}), \tag{8}$$

which concentrates mass uniformly on Σ. Substituting into (1) and applying the delta, the moment reduces immediately to:

$$C^{\mathrm{pgon}}_{mnp} = \iint_\Sigma \phi_{mnp}(x,y,z)\ \mathrm{d}S, \tag{9}$$

a surface integral over the planar polygon.

Key observation. Unlike the polyhedron, there is no enclosed volume, so the Divergence Theorem is *not* used. Instead, we triangulate Σ and parameterize each triangle directly, using the same simplex machinery developed for polyhedron faces in Section 8.

4.4 Polyhedron model

Let $\Omega \subset \mathbb{R}^3$ be the solid interior of a polyhedron. We use the indicator function:

$$f_{\text{poly}}(x,y,z) = \begin{cases} 1, & (x,y,z) \in \Omega, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

Moments are then volume integrals over Ω.

5 Case group 1: Point moments (3 cases), fully derived

Starting from (1) and substituting (6), and applying the sifting property of the Dirac delta three times:

$$C^{\text{pt}}_{mnp} = \iiint \delta(x-x_0)\delta(y-y_0)\delta(z-z_0)\,\phi_{mnp}(x,y,z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = \phi_{mnp}(x_0,y_0,z_0).$$

$$C^{\text{pt}}_{mnp} = \phi_{mnp}(x_0,y_0,z_0). \tag{11}$$

Substituting each basis:

$$C^{\text{pt,geo}}_{mnp} = x_0^m\, y_0^n\, z_0^p, \tag{12}$$

$$C^{\text{pt,F}}_{mnp} = e^{-\mathrm{j}\pi(mx_0+ny_0+pz_0)}, \tag{13}$$

$$C^{\text{pt,L}}_{mnp} = P_m(x_0)\,P_n(y_0)\,P_p(z_0). \tag{14}$$

6 Case group 2: Line segment moments (3 cases), fully derived

6.1 Step 1 — reduce the 3D integral to a 1D integral

Substitute (7) into (1) and swap the order of integration (justified by the bounded, well-behaved integrand):

$$C^{\text{seg}}_{mnp} = \int_0^1 \underbrace{\iiint \delta(x-x(t))\delta(y-y(t))\delta(z-z(t))\phi_{mnp}(x,y,z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z}_{=\phi_{mnp}(x(t),y(t),z(t))}\; L\,\mathrm{d}t.$$

$$C^{\text{seg}}_{mnp} = L\int_0^1 \phi_{mnp}(a_x+td_x,\; a_y+td_y,\; a_z+td_z)\,\mathrm{d}t. \tag{15}$$

This master formula is the starting point for all three segment bases.

6.2 Segment geometric moments: step-by-step to a finite sum

Plug $\phi^{\text{geo}}_{mnp} = x^m y^n z^p$ into (15):

$$C^{\text{seg,geo}}_{mnp} = L \int_0^1 (a_x + td_x)^m (a_y + td_y)^n (a_z + td_z)^p \, \mathrm{d}t.$$

Step A. Expand each factor by the binomial theorem:

$$(a_x + td_x)^m = \sum_{i=0}^{m} \binom{m}{i} a_x^{m-i} d_x^i \, t^i,$$
$$(a_y + td_y)^n = \sum_{j=0}^{n} \binom{n}{j} a_y^{n-j} d_y^j \, t^j,$$
$$(a_z + td_z)^p = \sum_{k=0}^{p} \binom{p}{k} a_z^{p-k} d_z^k \, t^k.$$

Step B. Multiply and collect powers of t:

$$(a_x + td_x)^m (a_y + td_y)^n (a_z + td_z)^p = \sum_{i=0}^{m} \sum_{j=0}^{n} \sum_{k=0}^{p} \binom{m}{i} \binom{n}{j} \binom{p}{k} a_x^{m-i} a_y^{n-j} a_z^{p-k} \, d_x^i d_y^j d_z^k \, t^{i+j+k}.$$

Step C. Integrate term-by-term using $\displaystyle\int_0^1 t^{i+j+k} \, \mathrm{d}t = \frac{1}{i+j+k+1}$:

$$C^{\text{seg,geo}}_{mnp} = L \sum_{i=0}^{m} \sum_{j=0}^{n} \sum_{k=0}^{p} \binom{m}{i} \binom{n}{j} \binom{p}{k} \frac{a_x^{m-i} \, a_y^{n-j} \, a_z^{p-k} \, d_x^i \, d_y^j \, d_z^k}{i+j+k+1}. \tag{16}$$

Sanity Check 6.1. When $m = n = p = 0$, only the $i = j = k = 0$ term survives and the formula gives $C_{000} = L$, the segment length. This is correct: integrating 1 over the segment should return its length.

6.3 Segment Fourier moments: step-by-step to a closed form

Plug $\phi^{\text{F}}_{mnp} = e^{-\mathrm{j}\pi(mx+ny+pz)}$ into (15):

$$C^{\text{seg,F}}_{mnp} = L \int_0^1 e^{-\mathrm{j}\pi[m(a_x+td_x)+n(a_y+td_y)+p(a_z+td_z)]} \, \mathrm{d}t.$$

Step A. Separate constant and t-dependent parts of the phase:

$$m(a_x + td_x) + n(a_y + td_y) + p(a_z + td_z) = \underbrace{(ma_x + na_y + pa_z)}_{\text{constant}} + t \underbrace{(md_x + nd_y + pd_z)}_{\text{slope}}.$$

Step B. Factor out the constant exponential. Define:

$$\alpha = \pi(md_x + nd_y + pd_z), \qquad \beta_0 = \pi(ma_x + na_y + pa_z).$$

Then:

$$C_{mnp}^{\mathrm{seg,F}} = L\, e^{-\mathrm{j}\beta_0} \int_0^1 e^{-\mathrm{j}\alpha t}\, \mathrm{d}t.$$

Step C. Evaluate the standard exponential integral:

$$\int_0^1 e^{-\mathrm{j}\alpha t}\, \mathrm{d}t = \begin{cases} \dfrac{1 - e^{-\mathrm{j}\alpha}}{\mathrm{j}\alpha}, & \alpha \neq 0, \\ 1, & \alpha = 0. \end{cases}$$

$$C_{mnp}^{\mathrm{seg,F}} = L\, e^{-\mathrm{j}\beta_0} \begin{cases} \dfrac{1 - e^{-\mathrm{j}\alpha}}{\mathrm{j}\alpha}, & \alpha \neq 0, \\ 1, & \alpha = 0, \end{cases} \tag{17}$$

where $\alpha = \pi(md_x + nd_y + pd_z)$ and $\beta_0 = \pi(ma_x + na_y + pa_z)$.

The factor $(1 - e^{-\mathrm{j}\alpha})/(\mathrm{j}\alpha)$ is a sinc-like complex function that attenuates high spatial frequencies, just as in the 2D case.

6.4 Segment Legendre moments: step-by-step to a finite sum

Plug $\phi_{mnp}^{\mathrm{L}} = P_m(x)P_n(y)P_p(z)$ into (15):

$$C_{mnp}^{\mathrm{seg,L}} = L \int_0^1 P_m(x(t))\, P_n(y(t))\, P_p(z(t))\, \mathrm{d}t.$$

Step A. Expand each Legendre polynomial into monomials using (5):

$$P_m(x(t)) = \sum_{r=0}^{\lfloor m/2 \rfloor} A_{m,r}\, (x(t))^{m-2r}, \quad P_n(y(t)) = \sum_{s=0}^{\lfloor n/2 \rfloor} A_{n,s}\, (y(t))^{n-2s}, \quad P_p(z(t)) = \sum_{u=0}^{\lfloor p/2 \rfloor} A_{p,u}\, (z(t))^{p-2u}.$$

So:

$$C_{mnp}^{\mathrm{seg,L}} = L \sum_{r,s,u} A_{m,r} A_{n,s} A_{p,u} \int_0^1 (x(t))^{m-2r} (y(t))^{n-2s} (z(t))^{p-2u}\, \mathrm{d}t.$$

Step B. Each remaining integral is exactly the geometric segment integral with exponents $(m-2r,\, n-2s,\, p-2u)$, computed by triple binomial expansion and term-by-term integration (identical to Section 6.2, now with three indices):

$$\int_0^1 (x(t))^{q_1} (y(t))^{q_2} (z(t))^{q_3}\, \mathrm{d}t = \sum_{i=0}^{q_1} \sum_{j=0}^{q_2} \sum_{k=0}^{q_3} \binom{q_1}{i} \binom{q_2}{j} \binom{q_3}{k} \frac{a_x^{q_1-i} a_y^{q_2-j} a_z^{q_3-k}\, d_x^i d_y^j d_z^k}{i+j+k+1}.$$

$$C_{mnp}^{\text{seg,L}} = L \sum_{r=0}^{\lfloor m/2 \rfloor} \sum_{s=0}^{\lfloor n/2 \rfloor} \sum_{u=0}^{\lfloor p/2 \rfloor} A_{m,r} A_{n,s} A_{p,u} \times \sum_{i=0}^{m-2r} \sum_{j=0}^{n-2s} \sum_{k=0}^{p-2u} \binom{m-2r}{i} \binom{n-2s}{j} \binom{p-2u}{k} \frac{a_x^{(m-2r)-i}\, a_y^{(n-2s)-j}\, a_z^{(p-2u)-k}\, d_x^i\, d_y^j\, d_z^k}{i+j+k+1}. \tag{18}$$

No integrals remain; every sum is finite.

7 Case group 4: Polygon moments (3 cases), fully derived

We now derive all three moment types for a flat polygon Σ embedded in $\mathbb{R}^3$. The moment is the surface integral (9).

7.1 Triangulation and parameterization

Any planar polygon Σ can be decomposed into T non-overlapping triangles:

$$\iint_{\Sigma} \phi_{mnp}\ \mathrm{d}S = \sum_{\tau=1}^{T} \iint_{\text{triangle } \tau} \phi_{mnp}\ \mathrm{d}S.$$

For each triangle τ with CCW vertices $\mathbf{a} = (x_0, y_0, z_0)$, $\mathbf{b} = (x_1, y_1, z_1)$, $\mathbf{c} = (x_2, y_2, z_2)$ (ordered CCW when viewed from the positive-normal side), define edge vectors:

$$\mathbf{e}_1 = \mathbf{b} - \mathbf{a} = (\Delta x_1, \Delta y_1, \Delta z_1), \qquad \mathbf{e}_2 = \mathbf{c} - \mathbf{a} = (\Delta x_2, \Delta y_2, \Delta z_2).$$

Parameterize the triangle by $(s, t) \in \mathcal{T} = \{s \geq 0,\ t \geq 0,\ s + t \leq 1\}$:

$$x(s,t) = x_0 + s\Delta x_1 + t\Delta x_2, \quad y(s,t) = y_0 + s\Delta y_1 + t\Delta y_2, \quad z(s,t) = z_0 + s\Delta z_1 + t\Delta z_2.$$

The area element is:

$$\mathrm{d}S = \|\mathbf{e}_1 \times \mathbf{e}_2\|\ \mathrm{d}s\ \mathrm{d}t = 2A_\tau\ \mathrm{d}s\ \mathrm{d}t,$$

where $A_\tau = \frac{1}{2}\|\mathbf{e}_1 \times \mathbf{e}_2\|$ is the triangle area.

Fundamental simplex integral (used throughout):

$$\iint_{\mathcal{T}} s^i\, t^j\ \mathrm{d}s\ \mathrm{d}t = \frac{i!\, j!}{(i+j+2)!}. \tag{19}$$

7.2 Polygon geometric moments: step-by-step to a finite sum

Substitute $\phi_{mnp}^{\text{geo}} = x^m y^n z^p$ into (9) and split over triangles:

$$C_{mnp}^{\text{pgon,geo}} = \sum_{\tau=1}^{T} 2A_\tau \iint_{\mathcal{T}} \big(x_0 + s\Delta x_1 + t\Delta x_2\big)^m \big(y_0 + s\Delta y_1 + t\Delta y_2\big)^n \big(z_0 + s\Delta z_1 + t\Delta z_2\big)^p\ \mathrm{d}s\ \mathrm{d}t.$$

Step A. Apply the binomial theorem twice to each factor. For a generic factor $(u_0 + s\delta_1 + t\delta_2)^q$, first group the s-term:

$$(u_0 + s\delta_1 + t\delta_2)^q = \sum_{i=0}^{q} \binom{q}{i} (u_0 + t\delta_2)^{q-i} (\delta_1)^i \, s^i,$$

then expand $(u_0 + t\delta_2)^{q-i}$:

$$(u_0 + t\delta_2)^{q-i} = \sum_{k=0}^{q-i} \binom{q-i}{k} u_0^{q-i-k} \, \delta_2^k \, t^k.$$

Combining:

$$(u_0 + s\delta_1 + t\delta_2)^q = \sum_{i=0}^{q} \sum_{k=0}^{q-i} \underbrace{\binom{q}{i}\binom{q-i}{k}}_{B^q_{i,k}} u_0^{q-i-k} \, \delta_1^i \, \delta_2^k \, s^i t^k.$$

Step B. Multiply the expansions for x, y, z and collect powers: the s-power is $I = i_x + i_y + i_z$ and the t-power is $K = k_x + k_y + k_z$.
Step C. Integrate term-by-term using (19):

$$C^{\text{pgon,geo}}_{mnp} = \sum_{\tau=1}^{T} 2A_\tau \sum_{\substack{0\le i_x,\, i_x+k_x\le m \\ 0\le i_y,\, i_y+k_y\le n \\ 0\le i_z,\, i_z+k_z\le p}} B^m_{i_x,k_x} \, B^n_{i_y,k_y} \, B^p_{i_z,k_z}$$
$$\times \, x_0^{m-i_x-k_x} \, y_0^{n-i_y-k_y} \, z_0^{p-i_z-k_z} \, \Delta x_1^{i_x} \Delta x_2^{k_x} \, \Delta y_1^{i_y} \Delta y_2^{k_y} \, \Delta z_1^{i_z} \Delta z_2^{k_z} \, \frac{(i_x+i_y+i_z)!\,(k_x+k_y+k_z)!}{(i_x+i_y+i_z+k_x+k_y+k_z+2)!}, \tag{20}$$

where $B^q_{i,k} = \binom{q}{i}\binom{q-i}{k}$.

Sanity Check 7.1. When $m = n = p = 0$: only the all-zero index term survives. The formula gives $C^{\text{pgon,geo}}_{000} = \sum_\tau 2A_\tau \cdot \frac{0!\,0!}{2!} = \sum_\tau 2A_\tau \cdot \frac{1}{2} = \sum_\tau A_\tau$, the total area of Σ. ✓

7.3 Polygon Fourier moments: step-by-step to a closed form

Substitute $\phi^{\text{F}}_{mnp} = e^{-\mathrm{j}\pi(mx+ny+pz)}$ into (9) and split over triangles:

$$C^{\text{pgon,F}}_{mnp} = \sum_{\tau=1}^{T} 2A_\tau \iint_{\mathcal{T}} e^{-\mathrm{j}\pi(mx(s,t)+ny(s,t)+pz(s,t))} \, \mathrm{d}s \, \mathrm{d}t.$$

Step A. The phase is linear in (s,t):

$$mx(s,t) + ny(s,t) + pz(s,t) = \frac{\beta_\tau}{\pi} + s\frac{\alpha_\tau}{\pi} + t\frac{\mu_\tau}{\pi},$$

with per-triangle constants:

$$\beta_\tau = \pi(mx_0 + ny_0 + pz_0), \quad \alpha_\tau = \pi(m\Delta x_1 + n\Delta y_1 + p\Delta z_1), \quad \mu_\tau = \pi(m\Delta x_2 + n\Delta y_2 + p\Delta z_2).$$

Step B. Factor out the constant phase:

$$\iint_{\mathcal{T}} e^{-\mathrm{j}\pi(mx+ny+pz)}\ \mathrm{d}s\ \mathrm{d}t = e^{-\mathrm{j}\beta_\tau} \iint_{\mathcal{T}} e^{-\mathrm{j}(\alpha_\tau s+\mu_\tau t)}\ \mathrm{d}s\ \mathrm{d}t\ =\ e^{-\mathrm{j}\beta_\tau}\ \mathcal{I}(\alpha_\tau, \mu_\tau),$$

where $\mathcal{I}(\alpha, \mu)$ is the 2D complex exponential simplex integral. It is evaluated by integrating over $s \in [0, 1-t]$ first, then over $t \in [0, 1]$:

$$\mathcal{I}(\alpha,\mu) = \begin{cases} \dfrac{1}{\mathrm{j}\alpha}\left[\dfrac{1-e^{-\mathrm{j}\mu}}{\mathrm{j}\mu} - \dfrac{e^{-\mathrm{j}\alpha}(1-e^{-\mathrm{j}(\mu-\alpha)})}{\mathrm{j}(\mu-\alpha)}\right], & \alpha \neq 0,\ \mu \neq 0,\ \alpha \neq \mu, \\ \dfrac{1-e^{-\mathrm{j}\alpha}(1-\mathrm{j}\alpha)-1}{\alpha^2}, & \alpha = \mu \neq 0, \\ \dfrac{1-e^{-\mathrm{j}\mu}-\mathrm{j}\mu e^{-\mathrm{j}\mu}}{-\mu^2}, & \alpha = 0,\ \mu \neq 0, \\ \frac{1}{2}, & \alpha = \mu = 0. \end{cases} \tag{21}$$

$$C_{mnp}^{\mathrm{pgon,F}} = \sum_{\tau=1}^{T} 2A_\tau\ e^{-\mathrm{j}\beta_\tau}\ \mathcal{I}(\alpha_\tau, \mu_\tau). \tag{22}$$

Contrast with the polyhedron Fourier case (31)**.** For the polyhedron each face integral is premultiplied by the dot product $(mn_{x,f} + nn_{y,f} + pn_{z,f})/(-\mathrm{j}\pi(m^2+n^2+p^2))$, which arises from the Divergence Theorem construction. For the polygon no such factor appears: we integrate ϕ directly over the surface, so only the area $2A_\tau$, the phase offset $e^{-\mathrm{j}\beta_\tau}$, and the simplex integral $\mathcal{I}$ enter.

Sanity Check 7.2. When $m = n = p = 0$: $\alpha_\tau = \mu_\tau = \beta_\tau = 0$, so $\mathcal{I}(0,0) = \frac{1}{2}$ and $e^0 = 1$. The formula gives $\sum_\tau 2A_\tau \cdot \frac{1}{2} = \sum_\tau A_\tau$. ✓

7.4 Polygon Legendre moments: reduction to geometric moments

$$C_{mnp}^{\mathrm{pgon,L}} = \iint_{\Sigma} P_m(x)\, P_n(y)\, P_p(z)\ \mathrm{d}S.$$

Step 1. Expand each Legendre polynomial using (5):

$$P_m(x)P_n(y)P_p(z) = \sum_{r,s,u} A_{m,r}\, A_{n,s}\, A_{p,u}\, x^{m-2r}\, y^{n-2s}\, z^{p-2u},$$

where the sums run $r = 0, \ldots, \lfloor m/2 \rfloor$, $s = 0, \ldots, \lfloor n/2 \rfloor$, $u = 0, \ldots, \lfloor p/2 \rfloor$.
Step 2. Swap sum and integral; the remaining integrals are geometric polygon moments:

$$C^{\text{pgon,L}}_{mnp} = \sum_{r=0}^{\lfloor m/2 \rfloor} \sum_{s=0}^{\lfloor n/2 \rfloor} \sum_{u=0}^{\lfloor p/2 \rfloor} A_{m,r}\, A_{n,s}\, A_{p,u}\, C^{\text{pgon,geo}}_{(m-2r),(n-2s),(p-2u)}, \tag{23}$$

where each $C^{\text{pgon,geo}}_{pqr}$ is given by (20).

The structure mirrors the polyhedron Legendre reduction (32) exactly, but calls the polygon geometric formula instead of the polyhedron one.

8 Case group 5: Polyhedron moments via the Divergence Theorem

Polyhedron moments are volume integrals:

$$C^{\text{poly}}_{mnp} = \iiint_{\Omega} \phi_{mnp}(x, y, z)\, \mathrm{d}V.$$

A direct approach would require integration over a complex 3D region. Instead we use a key vector calculus identity to convert volume integrals into surface integrals.

8.1 The Divergence Theorem: the version we use

$$\iiint_{\Omega} \big(\nabla \cdot \mathbf{F}\big)\, \mathrm{d}V = \oiint_{\partial\Omega} \mathbf{F} \cdot \hat{\mathbf{n}}\ \mathrm{d}S. \tag{24}$$

Here:

- Ω is the polyhedron interior,
- $\partial\Omega$ is its boundary (the faces),
- $\mathbf{F} = (F_x, F_y, F_z)$ is a smooth vector field,
- $\hat{\mathbf{n}}$ is the outward unit normal on the boundary,
- $\mathrm{d}S$ is the surface area element.

The key trick. If we can find $\mathbf{F}$ such that $\nabla \cdot \mathbf{F} = \phi_{mnp}$, then the left side of (24) becomes our moment:

$$C^{\text{poly}}_{mnp} = \oiint_{\partial\Omega} \mathbf{F} \cdot \hat{\mathbf{n}}\ \mathrm{d}S.$$

The problem reduces to: *find a vector field whose divergence equals the basis function.*

8.2 How a polyhedral surface integral becomes a sum over triangular faces

Any polyhedral surface $\partial\Omega$ is a union of flat polygonal faces. We triangulate each face so that the boundary decomposes into triangles:

$$\oiint_{\partial\Omega} (\cdot)\ \mathrm{d}S = \sum_{f=1}^{F} \iint_{\text{face } f} (\cdot)\ \mathrm{d}S.$$

If we can evaluate the integral on a single triangle, we can sum over all faces.

8.3 Parameterizing one triangular face and its outward normal

Consider a triangle with CCW vertices (as seen from outside) $\mathbf{a} = (x_0, y_0, z_0)$, $\mathbf{b} = (x_1, y_1, z_1)$, $\mathbf{c} = (x_2, y_2, z_2)$. Define the two edge vectors:

$$\mathbf{e}_1 = \mathbf{b} - \mathbf{a} = (\Delta x_1, \Delta y_1, \Delta z_1), \qquad \mathbf{e}_2 = \mathbf{c} - \mathbf{a} = (\Delta x_2, \Delta y_2, \Delta z_2).$$

Parameterize the triangle by $(s, t) \in \mathcal{T}$, where $\mathcal{T} = \{(s, t) : s \geq 0,\ t \geq 0,\ s + t \leq 1\}$:

$$x(s,t) = x_0 + s\Delta x_1 + t\Delta x_2, \quad y(s,t) = y_0 + s\Delta y_1 + t\Delta y_2, \quad z(s,t) = z_0 + s\Delta z_1 + t\Delta z_2. \tag{25}$$

The outward normal and area element come from the cross product:

$$\mathbf{e}_1 \times \mathbf{e}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \Delta x_1 & \Delta y_1 & \Delta z_1 \\ \Delta x_2 & \Delta y_2 & \Delta z_2 \end{vmatrix} = (n'_x, n'_y, n'_z),$$

where $n'_x = \Delta y_1 \Delta z_2 - \Delta z_1 \Delta y_2$, $n'_y = \Delta z_1 \Delta x_2 - \Delta x_1 \Delta z_2$, $n'_z = \Delta x_1 \Delta y_2 - \Delta y_1 \Delta x_2$.

The face area and unit outward normal are:

$$A = \tfrac{1}{2} \left\| \mathbf{e}_1 \times \mathbf{e}_2 \right\|, \qquad \hat{\mathbf{n}} = \frac{\mathbf{e}_1 \times \mathbf{e}_2}{\|\mathbf{e}_1 \times \mathbf{e}_2\|} = (n_x, n_y, n_z).$$

The surface element transforms as:

$$\mathrm{d}S = \|\mathbf{e}_1 \times \mathbf{e}_2\| \ \mathrm{d}s\ \mathrm{d}t = 2A\ \mathrm{d}s\ \mathrm{d}t.$$

Key triangle domain integral. The fundamental integral over the standard simplex $\mathcal{T}$ is:

$$\iint_{\mathcal{T}} s^i\, t^j\ \mathrm{d}s\ \mathrm{d}t = \frac{i!\ j!}{(i + j + 2)!}. \tag{26}$$

This is the 3D analogue of $\int_0^1 t^r\, \mathrm{d}t = 1/(r+1)$ used in 2D edge integrals. It replaces all face integrals, just as the 1D monomial integral replaced all edge integrals in the polygon case.

Sanity Check 8.1. For $i = j = 0$: $\iint_{\mathcal{T}} \mathrm{d}s\ \mathrm{d}t = \frac{1}{2}$, which is the area of the unit simplex. Combined with the $2A$ factor from $\mathrm{d}S$, this correctly gives a face area contribution of A.

8.4 Polygon geometric moments: very detailed derivation to a face formula

We want:

$$C_{mnp}^{\text{poly,geo}} = \iiint_{\Omega} x^m y^n z^p\, \mathrm{d}V.$$

Step 1: choose a vector field. We want $\nabla \cdot \mathbf{F} = x^m y^n z^p$. Put everything in F_x, and set $F_y = F_z = 0$:

$$F_x = \frac{x^{m+1} y^n z^p}{m+1}, \qquad F_y = 0, \qquad F_z = 0.$$

Then $\partial F_x / \partial x = x^m y^n z^p$ as required.

Step 2: apply the Divergence Theorem. Since $F_y = F_z = 0$:

$$C_{mnp}^{\text{poly,geo}} = \oiint_{\partial\Omega} F_x n_x\ \mathrm{d}S = \oiint_{\partial\Omega} \frac{x^{m+1} y^n z^p}{m+1}\, n_x\ \mathrm{d}S. \tag{27}$$

Step 3: split into faces. On each triangular face, n_x is constant (the face is flat), so:

$$C_{mnp}^{\text{poly,geo}} = \sum_{f=1}^{F} \frac{n_{x,f}}{m+1} \iint_{\text{face } f} x^{m+1} y^n z^p \ \mathrm{d}S.$$

Step 4: parameterize one face and reduce to a 2D integral. On face f with vertices $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and parameterization (25):

$$\iint_{\text{face}} x^{m+1} y^n z^p \ \mathrm{d}S = 2A_f \iint_{\mathcal{T}} (x_0+s\Delta x_1+t\Delta x_2)^{m+1} (y_0+s\Delta y_1+t\Delta y_2)^n (z_0+s\Delta z_1+t\Delta z_2)^p \, \mathrm{d}s \ \mathrm{d}t.$$

Step 5: expand and integrate term-by-term. Apply the multinomial theorem (as three binomials in sequence) to each factor. For the exponent q factor with base u_0 and edge increments δ_1, δ_2:

$$(u_0 + s\delta_1 + t\delta_2)^q = \sum_{i=0}^{q} \sum_{k=0}^{q-i} \binom{q}{i} \binom{q-i}{k} u_0^{q-i-k} \, \delta_1^i \, \delta_2^k \, s^i t^k.$$

Multiplying the three such expansions (for x, y, z) and collecting powers of s and t, then integrating using (26):

$$C_{mnp}^{\text{poly,geo}} = \sum_{f=1}^{F} \frac{2A_f \, n_{x,f}}{m+1} \sum_{\substack{i_1+i_2 \le m+1 \\ j_1+j_2 \le n \\ k_1+k_2 \le p}} B_{i_1,i_2}^{m+1} \, B_{j_1,j_2}^{n} \, B_{k_1,k_2}^{p}$$
$$\times \, x_0^{m+1-i_1-i_2} \, y_0^{n-j_1-j_2} \, z_0^{p-k_1-k_2} \, \Delta x_1^{i_1} \Delta x_2^{i_2} \, \Delta y_1^{j_1} \Delta y_2^{j_2} \, \Delta z_1^{k_1} \Delta z_2^{k_2} \, \frac{(i_1+j_1+k_1)! \, (i_2+j_2+k_2)!}{(i_1+j_1+k_1+i_2+j_2+k_2+2)!}, \tag{28}$$

where the coefficient for exponent q is:

$$B_{i,k}^{q} = \binom{q}{i} \binom{q-i}{k}, \qquad 0 \le i, \ 0 \le k, \ i+k \le q.$$

This is an explicit finite closed form. All sums have at most $(m+2)(m+2)(n+1)(n+1)(p+1)(p+1)$ terms total.

8.5 Polyhedron Fourier moments: very detailed derivation to a face formula

We want:

$$C_{mnp}^{\text{poly,F}} = \iiint_{\Omega} e^{-\mathrm{j}\pi(mx+ny+pz)} \, \mathrm{d}V.$$

Step 1: choose a vector field. Because derivatives of $e^{-\mathrm{j}\pi(mx+ny+pz)}$ reproduce the same exponential, we try $\mathbf{F} = \gamma \, (m, n, p)^T e^{-\mathrm{j}\pi(mx+ny+pz)}$. Computing the divergence:

$$\nabla \cdot \mathbf{F} = \gamma(-\mathrm{j}\pi m^2 - \mathrm{j}\pi n^2 - \mathrm{j}\pi p^2) \, e^{-\mathrm{j}\pi(mx+ny+pz)} = \gamma(-\mathrm{j}\pi)(m^2+n^2+p^2) \, e^{-\mathrm{j}\pi(mx+ny+pz)}.$$

Setting this equal to $e^{-\mathrm{j}\pi(mx+ny+pz)}$ gives:

$$\gamma = \frac{1}{-\mathrm{j}\pi(m^2+n^2+p^2)}.$$

So:

$$\mathbf{F}(x,y,z) = \frac{1}{-\mathrm{j}\pi(m^2+n^2+p^2)} \begin{pmatrix} m \\ n \\ p \end{pmatrix} e^{-\mathrm{j}\pi(mx+ny+pz)}.$$

Step 2: apply the Divergence Theorem.

$$C_{mnp}^{\mathrm{poly,F}} = \oiint_{\partial\Omega} \mathbf{F}\cdot\hat{\mathbf{n}}\ \mathrm{d}S = \frac{1}{-\mathrm{j}\pi(m^2+n^2+p^2)} \oiint_{\partial\Omega} (mn_x+nn_y+pn_z)\, e^{-\mathrm{j}\pi(mx+ny+pz)}\ \mathrm{d}S.$$

Step 3: reduce to face integrals. On face f, $\hat{\mathbf{n}}_f = (n_{x,f}, n_{y,f}, n_{z,f})$ is constant:

$$C_{mnp}^{\mathrm{poly,F}} = \frac{1}{-\mathrm{j}\pi(m^2+n^2+p^2)} \sum_{f=1}^{F} (mn_{x,f}+nn_{y,f}+pn_{z,f}) \iint_{\text{face } f} e^{-\mathrm{j}\pi(mx+ny+pz)}\ \mathrm{d}S.$$

Step 4: evaluate the face exponential integral. With parameterization (25), the phase is:

$$mx(s,t)+ny(s,t)+pz(s,t) = (mx_0+ny_0+pz_0) + s(m\Delta x_1+n\Delta y_1+p\Delta z_1) + t(m\Delta x_2+n\Delta y_2+p\Delta z_2).$$

Define per-face constants:

$$\beta_f = \pi(mx_{0,f}+ny_{0,f}+pz_{0,f}), \quad \alpha_f = \pi(m\Delta x_{1,f}+n\Delta y_{1,f}+p\Delta z_{1,f}), \quad \mu_f = \pi(m\Delta x_{2,f}+n\Delta y_{2,f}+p\Delta z_{2,f}).$$

Then:

$$\iint_{\text{face}} e^{-\mathrm{j}\pi(mx+ny+pz)}\ \mathrm{d}S = 2A_f\, e^{-\mathrm{j}\beta_f} \iint_{\mathcal{T}} e^{-\mathrm{j}(\alpha_f s+\mu_f t)}\ \mathrm{d}s\ \mathrm{d}t.$$

Step 5: evaluate the 2D exponential simplex integral. The general result for the exponential integral over the unit simplex is:

$$\iint_{\mathcal{T}} e^{-\mathrm{j}(\alpha s+\mu t)}\ \mathrm{d}s\ \mathrm{d}t = \begin{cases} \dfrac{e^{-\mathrm{j}\alpha}-e^{-\mathrm{j}\mu}}{(\mathrm{j})^2(\alpha-\mu)\mu} + \dfrac{1-e^{-\mathrm{j}\mu}}{(\mathrm{j})^2\mu(\alpha-\mu)\cdot(-1)} & \text{(general form, derived below)}, \\ \dfrac{1-e^{-\mathrm{j}\alpha}-\mathrm{j}\alpha e^{-\mathrm{j}\alpha}}{-\alpha^2} & \alpha=\mu\neq 0, \\ \frac{1}{2} & \alpha=\mu=0. \end{cases} \tag{29}$$

A cleaner closed form is obtained by integrating over s first (holding t fixed), then over t. For the inner s-integral:

$$\int_0^{1-t} e^{-\mathrm{j}(\alpha s+\mu t)}\ \mathrm{d}s = e^{-\mathrm{j}\mu t} \int_0^{1-t} e^{-\mathrm{j}\alpha s}\ \mathrm{d}s.$$

If $\alpha \neq 0$:

$$\int_0^{1-t} e^{-\mathrm{j}\alpha s}\ \mathrm{d}s = \frac{1-e^{-\mathrm{j}\alpha(1-t)}}{\mathrm{j}\alpha}.$$

Then integrating over t from 0 to 1 requires one more standard integral (exponential times polynomial), which yields:

$$\iint_{\mathcal{T}} e^{-\mathrm{j}(\alpha s+\mu t)}\ \mathrm{d}s\ \mathrm{d}t = \begin{cases} \dfrac{1}{\mathrm{j}\alpha}\left[\dfrac{1-e^{-\mathrm{j}\mu}}{\mathrm{j}\mu} - \dfrac{e^{-\mathrm{j}\alpha}(1-e^{-\mathrm{j}(\mu-\alpha)})}{\mathrm{j}(\mu-\alpha)}\right], & \alpha\neq 0,\ \mu\neq 0,\ \alpha\neq\mu, \\ \dfrac{1}{\mathrm{j}\alpha}\left[\dfrac{1-e^{-\mathrm{j}\alpha}}{\mathrm{j}\alpha} + \dfrac{e^{-\mathrm{j}\alpha}-1+\mathrm{j}\alpha e^{-\mathrm{j}\alpha}}{\alpha^2/(-1)}\right], & \alpha=\mu\neq 0, \\ \dfrac{1-e^{-\mathrm{j}\mu}-\mathrm{j}\mu}{\mu^2}, & \alpha=0,\ \mu\neq 0, \\ \frac{1}{2}, & \alpha=\mu=0. \end{cases} \tag{30}$$

We denote this function $\mathcal{I}(\alpha,\mu)$ for compactness.

$$C^{\mathrm{poly,F}}_{mnp} = \frac{1}{-\mathrm{j}\pi(m^2+n^2+p^2)}\sum_{f=1}^{F} 2A_f\,(mn_{x,f}+nn_{y,f}+pn_{z,f})\,e^{-\mathrm{j}\beta_f}\,\mathcal{I}(\alpha_f,\mu_f), \tag{31}$$

where $\mathcal{I}(\alpha,\mu)$ is given by (30), $\beta_f=\pi(mx_{0,f}+ny_{0,f}+pz_{0,f})$, $\alpha_f=\pi(m\Delta x_{1,f}+n\Delta y_{1,f}+p\Delta z_{1,f})$, $\mu_f=\pi(m\Delta x_{2,f}+n\Delta y_{2,f}+p\Delta z_{2,f})$.

8.6 Polyhedron Legendre moments: reduction to geometric moments

We want:

$$C^{\mathrm{poly,L}}_{mnp} = \iiint_{\Omega} P_m(x)\,P_n(y)\,P_p(z)\ \mathrm{d}V.$$

Step 1. Expand each Legendre polynomial into monomials using (5):

$$P_m(x)P_n(y)P_p(z) = \sum_{r=0}^{\lfloor m/2\rfloor}\sum_{s=0}^{\lfloor n/2\rfloor}\sum_{u=0}^{\lfloor p/2\rfloor} A_{m,r}\,A_{n,s}\,A_{p,u}\,x^{m-2r}\,y^{n-2s}\,z^{p-2u}.$$

Step 2. Swap the sum and integral (linearity):

$$C^{\mathrm{poly,L}}_{mnp} = \sum_{r,s,u} A_{m,r}\,A_{n,s}\,A_{p,u}\iiint_{\Omega} x^{m-2r}\,y^{n-2s}\,z^{p-2u}\ \mathrm{d}V.$$

The remaining integrals are geometric polyhedron moments:

$$\iiint_{\Omega} x^{m-2r}\,y^{n-2s}\,z^{p-2u}\ \mathrm{d}V = C^{\mathrm{poly,geo}}_{(m-2r),(n-2s),(p-2u)}.$$

$$C_{mnp}^{\text{poly,L}} = \sum_{r=0}^{\lfloor m/2 \rfloor} \sum_{s=0}^{\lfloor n/2 \rfloor} \sum_{u=0}^{\lfloor p/2 \rfloor} A_{m,r}\, A_{n,s}\, A_{p,u}\, C_{(m-2r),(n-2s),(p-2u)}^{\text{poly,geo}}. \tag{32}$$

Each $C_{pqr}^{\text{poly,geo}}$ is computed by (28), so Legendre polyhedron moments are fully closed form.

9 Examples (conceptual setup for Chunk 2)

9.1 Polygon example: a tilted quadrilateral in $\mathbb{R}^3$

Use four coplanar vertices forming a quadrilateral not parallel to any coordinate plane:

$$\mathbf{a} = (-0.3,\, -0.2,\, 0.1), \quad \mathbf{b} = (0.4,\, -0.1,\, 0.2), \quad \mathbf{c} = (0.3,\, 0.4,\, 0.15), \quad \mathbf{d} = (-0.2,\, 0.3,\, 0.05).$$

(Coplanarity should be verified; the normal $\mathbf{e}_1 \times \mathbf{e}_2$ is the same for both triangles of the fan triangulation.) Triangulate as $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ and $(\mathbf{a}, \mathbf{c}, \mathbf{d})$. Because the polygon is tilted, x, y, and z all vary across the surface, so moments will generally be nonzero for all (m, n, p).

9.2 Polyhedron example A: Cube centered at origin, side length 0.3

Half-side $a = 0.15$. The 8 vertices are $(\pm a, \pm a, \pm a)$. Six square faces, each triangulated into 2 right triangles, giving 12 triangles total. Counterclockwise winding (as seen from outside) must be maintained for correct normals.

Why the cube collapses. Because the cube is axis-aligned and symmetric, the interior is $\Omega = [-a, a]^3$ and all volume integrals factor:

$$\iiint_\Omega x^m y^n z^p \, \mathrm{d}V = \int_{-a}^{a} x^m \, \mathrm{d}x \int_{-a}^{a} y^n \, \mathrm{d}y \int_{-a}^{a} z^p \, \mathrm{d}z.$$

Each 1D integral is zero when the corresponding exponent is odd. So $C_{mnp}^{\text{cube,geo}} = 0$ if any of m, n, p is odd. This provides a strong numerical check.

9.3 Polyhedron example B: A less symmetric tetrahedron

Use vertices:

$$\mathbf{a} = (-0.3, -0.2, -0.1), \quad \mathbf{b} = (0.4, -0.1, 0.0), \quad \mathbf{c} = (0.2, 0.4, -0.05), \quad \mathbf{d} = (-0.1, 0.1, 0.5).$$

Four triangular faces (one per combination of three vertices), with outward normals computed by the cross product rule. This shape has no symmetry, so moments will generally be nonzero for all (m, n, p), making it a thorough test.

10 Summary of closed forms

Shape	Basis	Equation	Method
Point	Geometric	(12)	Dirac sifting
Point	Fourier	(13)	Dirac sifting
Point	Legendre	(14)	Dirac sifting
Segment	Geometric	(16)	1D param. + binomial
Segment	Fourier	(17)	1D param. + sinc
Segment	Legendre	(18)	reduce to geo
Polygon	Geometric	(20)	triangulate + simplex integral
Polygon	Fourier	(22)	triangulate + $\mathcal{I}(\alpha, \mu)$
Polygon	Legendre	(23)	reduce to (20)
Polyhedron	Geometric	(28)	Div. Thm. + simplex integral
Polyhedron	Fourier	(31)	Div. Thm. + $\mathcal{I}(\alpha, \mu)$
Polyhedron	Legendre	(32)	reduce to (28)

Relationship between polygon and polyhedron. The polygon (Case group 4) and the polyhedron (Case group 5) share all their triangle-level machinery — the parameterization (25), the simplex integral (26), and the function $\mathcal{I}(\alpha, \mu)$ — but differ in how the triangle contributions are assembled:

- **Polygon**: each triangle contributes $2A_\tau\, e^{-\mathrm{j}\beta_\tau}\mathcal{I}$ directly to the surface integral (no divergence theorem step).
- **Polyhedron**: each face contributes the same triangle integral, but pre-weighted by $(mn_x + nn_y + pn_z)/(-\mathrm{j}\pi(m^2 + n^2 + p^2))$ from the Divergence Theorem construction.

The principal structural differences between the 2D and 3D cases are:

1. Moments now carry three indices (m, n, p) instead of two (m, n).
2. The edge parameterization $t \in [0, 1]$ becomes a triangle parameterization $(s, t) \in \mathcal{T}$, and all binomial expansions acquire a third index.
3. The fundamental domain integral $\int_0^1 t^r \,\mathrm{d}t = 1/(r+1)$ becomes $\iint_{\mathcal{T}} s^i t^j \,\mathrm{d}s\,\mathrm{d}t = i!\,j!/(i+j+2)!$.
4. The Fourier face/triangle integral requires the 2D complex exponential simplex function $\mathcal{I}(\alpha, \mu)$ in place of the 1D sinc-like function.
5. The Green's theorem vector field trick extends to the 3D Divergence Theorem for polyhedra; flat polygons bypass this step entirely.

Everything else — the overall structure, the Legendre reduction, the per-triangle summation — is a direct and natural generalization of the 2D derivation. Chunk 2 will add numerical examples, per-triangle tables, and modular Python code reproducing all results.